

SUSHMITHA KATHERINE JAYARAJ

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Data Analyst and Engineer with 4+ years of experience building Python and SQL driven analytical tools and data models, including hands on work with loan level real estate and mortgage risk data. Experienced partnering with cross functional stakeholders across enterprise engagements to define scope, gather requirements, and deliver governed, quality controlled data solutions. Skilled in Tableau and Power BI for dashboarding, statistical analysis and data modeling, and data governance practices including schema validation, lineage, and automated quality testing.

Core Competencies: Python & SQL Development · Financial & Real Estate Data Modeling · Data Governance & Quality · Statistical Analysis · Stakeholder & Cross Functional Collaboration · BI Dashboarding · Git & CI/CD

EDUCATION

Master's in Data Analytics, Conc. Applied Machine Learning, Northeastern University
B.E. in Information Science Engineering, Visvesvaraya Technological University

Sep 2024 to Jun 2026
Aug 2016 to Aug 2020

TECHNICAL SKILLS

Languages & Query: Python · SQL (Advanced) · NumPy · Pandas · R Programming

Data Modeling & Warehousing: Snowflake · Dimensional Modeling · Star Schema Design · Google BigQuery (familiar)

Data Governance & Quality: dbt Tests · Schema Validation · Data Lineage · Metadata Documentation · Access Controls · Observability

BI & Visualization: Tableau (Advanced) · Power BI · DAX · Streamlit

Version Control & Deployment: Git · GitHub · GitHub Actions (CI/CD)

Cloud: AWS (S3, Glue, Athena) · Azure · GCP

ETL/ELT & Orchestration: Airflow · dbt (Cloud + Core) · AWS Glue (PySpark)

PROFESSIONAL EXPERIENCE

Data Engineer Intern | Dell Technologies | Franklin, MA

Jul 2025 to Dec 2025

- Analyzed large scale manufacturing datasets using SQL and Python, applying data manipulation and business analytics techniques to build dbt based data models that translated stakeholder requirements into structured, tested, and documented analytical datasets.
- Developed automated data quality and validation frameworks with observability checks across multiple source systems, triaging pipeline failures and ensuring accurate, repeatable reporting across different schemas and update cadences.
- Designed self service Power BI dashboards and visualizations surfacing manufacturing KPIs and operational trends, enabling cross functional teams to interpret data and act without engineering intervention.

Data Analyst | Deloitte | Hyderabad, India

Feb 2022 to Sep 2024

- Gathered, normalized, and validated multi source financial and operational data across 3 enterprise client engagements, partnering with senior technology and business stakeholders to develop measurement frameworks that supported leadership decision making and transaction related analysis.
- Built scalable ELT pipelines using Python and SQL processing 10M+ records per run, applying advanced business analytics and data manipulation methods to convert complex, multi source transactional data into curated, quality controlled analytical products.
- Designed and delivered Tableau and Power BI dashboards for business stakeholders, using data visualization and storytelling to translate large, dense datasets into clear, repeatable analyses that reduced time to insight and supported strategic decisions.

Risk Quality Analyst | Amazon | Bangalore, India

Sep 2020 to Feb 2022

- Investigated refund abuse patterns across large scale transactional records using statistical analysis and data mining techniques, identifying anomalies in accounting and payment data that informed enforcement decisions.
- Built end to end data pipelines on 100M+ operational records using Python and SQL, applying advanced analytics to enable data driven decisions on resource allocation, capacity planning, and operational performance.
- Owned full analytics infrastructure, including pipeline design, data quality checks, monitoring, and reporting, managing multiple priorities in a fast moving environment with messy, ambiguous data at scale.

PROJECTS

Freddie Mac Mortgage Risk Analytics | Snowflake · dbt · Python · Streamlit · Git

- Built an end to end data pipeline on 150,000 loans and 6.2M monthly real estate loan performance records, designing a dimensional star schema in dbt with 11 automated data quality tests enforcing key uniqueness and referential integrity.

AWS Serverless ELT Pipeline | AWS Glue (PySpark) · S3 · Athena · Python · Git

- Built a serverless, cloud deployed ELT pipeline ingesting 1.87GB of NOAA weather data (24,630 files) into S3, using Glue PySpark to convert CSV to partitioned, Snappy compressed Parquet, cutting Athena query costs by roughly 52x.

Olist E-Commerce Intelligence Platform | Databricks · Delta Lake · PySpark · MLflow

- Built a Databricks medallion architecture pipeline ingesting 100K+ e-commerce orders across 9 Delta Lake tables, validating and normalizing order, payment, and inventory adjacent data with schema checks at each layer.

DocBrief | React · TypeScript · FastAPI · Groq (Llama 3) · ChromaDB

- Built a clinical decision support RAG pipeline: retrieves relevant clinical guidelines from an 18-entry knowledge base across 8 medical domains using semantic vector search, then synthesizes an evidence-backed clinical assessment with inline citations, reducing hours of literature review to seconds.

CERTIFICATIONS

Alteryx Designer Core [Certificate](#) • Python Programming, HackerRank [Certificate](#) • SQL (Advanced), HackerRank [Certificate](#)